

Forecasting GDP Growth Rates Using Accounting Earnings: A Large Panel Microdata Approach

Management Science, 2026

Yumeng Cui, Yongmiao Hong, Naijing Huang, Yicheng Wang

Presented by Kang Guo

August 2, 2026

Overview

1. Introduction

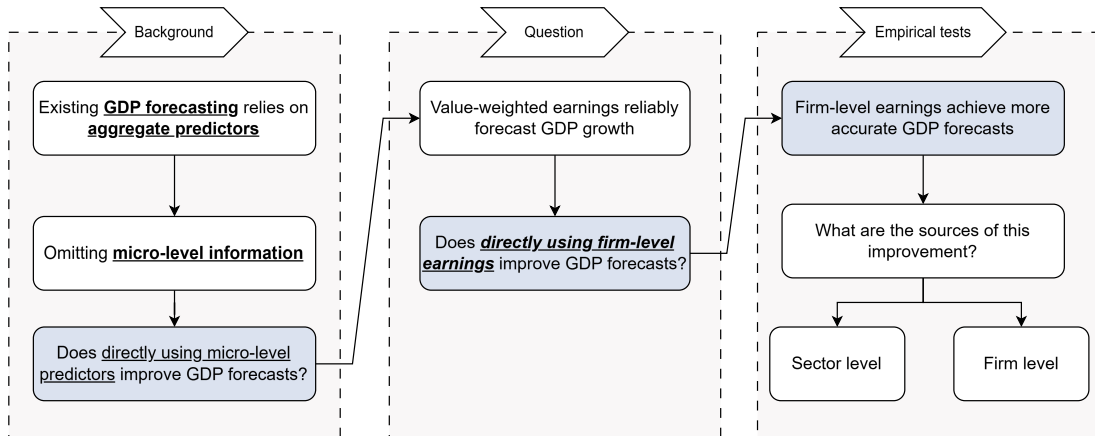
2. Design

3. Results

4. Replication

5. Ideal

Summary



Question

1. Does **directly using firm-level earnings** improve GDP forecasts?
 - 1.1 Lagged GDP growth
 - 1.2 Simple aggregate predictors
 - 1.3 Factor-based predictors
2. What are the sources of this improvement?
 - 2.1 Sector level
 - 2.2 Firm level

Motivation

- All existing GDP forecasting relies on **aggregated** predictors
 1. Lagged GDP growth
 2. Simple aggregate predictors: Mean, median, or dispersion
 3. Factor-based predictors: PCA (Abdalla et al., 2021, JME)
- **Challenge:** These aggregations inevitably **obscure micro-level information**
 - Omit information about microlevel heterogeneity and nonlinear interactions
- **Question:** Does directly using micro-level information improve GDP forecasts?
- Value-weighted average firm-level earnings changes are significant predictors of future GDP growth (Gaertner et al., 2020, RAS; Konchitchki and Patatoukas., 2014, JAE)
- **This paper:** Can directly using firm-level earnings better predict GDP growth?

Marginal Contributions

- **The Forecast of GDP Growth**

- **Prior work:** Rely on **aggregate economic and financial predictors**, such as earnings, firm's term spreads, stock returns, and interest rates (Gaertner et al., 2020, RAS; Ang et al., 2006, JE; Gilchrist and Zakrajsek, 2012, AER)
- **This paper:** **Directly using micro-level variables** (firm-level earnings) improves the prediction of GDP growth

Microforecasting Models

- **Direct forecasting equation** for GDP growth using firm-level earnings:

$$g_{t+h} = F_h(\mathbf{X}_t; \gamma) + \varepsilon_{t+h}, \quad t = 1, 2, \dots, T - h, \quad (1)$$

- **Target variable** (g_{t+h}): Cumulative real GDP growth rate from quarter t to $t + h$
- **Predictor vector** ($\mathbf{X}_t$): Consists of 3,005 features:
 - Accounting earnings from the 1,500 selected firms and their one-period lags
 - GDP growth rate g_t in period t and its four lags

Benchmark Model

- Use classic AR model as a benchmark

$$g_{t+h} = \mu_h + A(L)g_t + \varepsilon_{t+h}, \quad (2)$$

- Where $A(L)$ is a polynomial in lag operator L
- The lag order in $A(L)$ is determined by the Bayesian information criterion (BIC)

Data

- **Macroeconomic target:** Quarterly U.S. real GDP sourced from the FRED database
- **Target variable (g_{t+h}):** Cumulative real GDP growth rate:

$$g_{t+h} = 100 \times (\text{GDP}_{t+h} - \text{GDP}_t) / \text{GDP}_t \quad (3)$$

- **Firm-level predictor:** Compustat accounting earnings (quarterly **net income / sales**)
- **No look-ahead bias:** All predictors are available no later than the release of g_{t+h}
- **Sample period:** 1989Q2–2023Q4 (OOS: 2000Q1–2023Q4)

Firm-Level Earnings vs. lagged GDP Growth (AR Model)

1. Firm-level earnings provide **incremental predictive power over lagged GDP growth**
2. **Nonlinear algorithms outperform their linear counterparts**, especially for long-term forecasts

Microforecasting model	Forecast horizon				Average
	$h = 1$	$h = 2$	$h = 4$	$h = 8$	
adaLASSO	0.504**	0.893*	0.885*	0.895***	0.794
LASSO	0.503**	0.903**	0.874*	0.884***	0.791
Elastic Net	0.503**	0.887*	0.935*	0.898***	0.806
Ridge	0.552**	0.828**	0.898*	0.878*	0.789
RF	0.507**	0.733**	0.875*	0.768**	0.721
GBRT	0.674**	0.822**	0.843**	0.808*	0.787

Notes. This table reports the **MSE ratio of each microforecasting model relative to the AR benchmark for each forecast horizon**

Firm-Level Earnings vs. Simple Aggregate Predictors

- All prediction models include lagged GDP growth
- Firm-level earnings **outperform simple aggregate predictors**

Predictor	Forecast horizon			
	$h = 1$	$h = 2$	$h = 4$	$h = 8$
Panel A: Value-weighted aggregate accounting earnings (OLS)				
Aggregate	1.100	1.152	1.401	1.102
Panel B: SPF (Median consensus forecast; OLS)				
SPF	1.754	1.361	1.401	—
Panel C: FRED-QD database (166 macroeconomic predictors; RF)				
FRED-QD	0.939	1.197	1.043	0.845
FRED-QD + Firm-level earnings	0.508	0.731	0.882	0.771
Panel D: Individual accounting earnings (RF)				
Firm-level earnings	0.507	0.733	0.875	0.768

Notes. Out-of-sample MSE ratios relative to AR benchmark

Firm-Level Earnings vs. Factor-based Predictors

- Panel A: PCA aggregation of firm-level earnings
- Firm-level earnings **outperform factor-based predictors**

Model	Forecast horizon			
	$h = 1$	$h = 2$	$h = 4$	$h = 8$
Panel A: Principal component regression				
PCA	0.512	1.193	4.455	7.315
Panel B: Machine learning methods				
RF	0.507	0.733	0.875	0.768
Ridge	0.552	0.828	0.898	0.878

Notes. This table reports the MSE ratio of each model relative to the AR benchmark for each forecast horizon.

Sectoral Importance

- The three most significant sectors are **manufacturing, transportation, and finance**

Sector	Forecast horizon, %				Average, %
	$h = 1$	$h = 2$	$h = 4$	$h = 8$	
Manufacturing	19.21	24.70	35.43	35.63	28.74
Transportation	52.43	35.12	12.15	7.84	26.89
Finance	3.78	9.10	23.68	27.30	15.97
Services	7.14	7.38	15.03	15.75	11.32
Retail	5.17	6.09	2.53	4.25	4.51
AR term	7.52	7.36	0.07	0.23	3.80
Wholesale	3.01	5.93	3.34	2.05	3.58
Mining	1.09	1.73	4.43	4.78	3.01
Construction	0.14	0.29	1.56	1.19	0.79
Agriculture	0.36	2.06	0.38	0.12	0.73
Public administration	0.15	0.23	1.41	0.86	0.66

Notes. This table reports the detailed results of sectoral variable importance analysis based on RF

Firm-Level Variable Importance and Firm Characteristics

1. Firm age exhibits an **inverted U-shaped** relationship with forecasting importance.
2. Firms with **larger total assets, greater asset tangibility, and better financing conditions** tend to have higher forecasting importance

	(1) All h	(2) $h = 1$	(3) $h = 2,4$	(4) $h = 8$
Age	0.228*** (0.0149)	0.386*** (0.0334)	0.160*** (0.0187)	0.102*** (0.0311)
Age squared	-0.00146*** (0.000113)	-0.00226*** (0.000255)	-0.00109*** (0.000141)	-0.000739*** (0.000240)
Assets (log)	0.0651* (0.0337)	0.425*** (0.0794)	0.0397 (0.0425)	-0.248*** (0.0645)
Sales growth	-0.338 (0.272)	-0.114 (0.651)	-0.185 (0.339)	-0.928* (0.519)
Investment rate	2.66*** (0.601)	1.19 (1.51)	2.04*** (0.749)	4.72*** (1.09)
BM ratio	-0.363*** (0.0707)	-0.501*** (0.166)	-0.159* (0.0894)	-0.552*** (0.134)
Tobin's q	0.220*** (0.0338)	0.478*** (0.0857)	0.161*** (0.0422)	-0.0378 (0.0609)
Debt/assets	-1.96*** (0.265)	-5.71*** (0.654)	-1.22*** (0.333)	0.578 (0.486)
Dummy: paying dividend	2.74*** (0.165)	4.06*** (0.382)	3.09*** (0.212)	-0.0466 (0.311)
Tangible/assets	2.69*** (0.419)	7.24*** (0.972)	1.03* (0.532)	0.275 (0.797)
Observations	291,458	78,524	151,990	60,944
R^2	0.021	0.055	0.020	0.004
Time fixed effects	Yes	Yes	Yes	Yes
Sector fixed effects	Yes	Yes	Yes	Yes

Notes. Regressing firm-level relative variable importance on firm characteristics

Firm-Level Earnings vs. lagged GDP Growth (AR Model)

- Random Forest (RF) is the best-performing model in both the US and China

Microforecasting model	Original (US)					Replication (China)				
	$h = 1$	$h = 2$	$h = 4$	$h = 8$	Average	$h = 1$	$h = 2$	$h = 4$	$h = 8$	Average
adaLASSO	0.504**	0.893*	0.885*	0.895***	0.794	0.294*	0.483**	0.529**	0.688**	0.499
LASSO	0.503**	0.903**	0.874*	0.884***	0.791	0.294*	0.483**	0.534**	0.730**	0.510
Elastic Net	0.503**	0.887*	0.935*	0.898***	0.806	0.294*	0.483**	0.528**	0.758**	0.516
Ridge	0.552**	0.828**	0.898*	0.878*	0.789	0.258**	0.473**	0.550**	0.848**	0.532
RF	0.507**	0.733**	0.875*	0.768**	0.721	0.281**	0.490**	0.502**	0.653***	0.481
GBRT	0.674**	0.822**	0.843**	0.808*	0.787	0.282**	0.519**	0.455***	0.692**	0.487

Notes. This table reports the **MSE ratio of each microforecasting model relative to the AR benchmark for each forecast horizon**

Firm-Level Earnings vs. Simple Aggregate Predictors

- Firm-level heterogeneity critically improves GDP forecasts in both the US and China

	Original (US)				Replication (China)			
Predictor	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 1$	$h = 2$	$h = 4$	$h = 8$
Panel A: Value-weighted aggregate accounting earnings (OLS)								
Aggregate	1.100	1.152	1.401	1.102	0.861	1.030	1.241	1.271
Panel D: Individual accounting earnings (RF)								
Firm-level	0.507	0.733	0.875	0.768	0.281	0.490	0.502	0.653

Notes. Out-of-sample MSE ratios relative to AR benchmark

Firm-Level Earnings vs. Factor-based Predictors

Model	Original (US)				Replication (China)			
	$h = 1$	$h = 2$	$h = 4$	$h = 8$	$h = 1$	$h = 2$	$h = 4$	$h = 8$
Panel A: Principal component regression								
PCA	0.512	1.193	4.455	7.315	1.158	1.632	2.966	3.162
Panel B: Machine learning methods								
RF	0.507	0.733	0.875	0.768	0.281	0.490	0.502	0.653
Ridge	0.552	0.828	0.898	0.878	0.258	0.473	0.550	0.848

Notes. This table reports the MSE ratio of each model relative to the AR benchmark for each forecast horizon.

Sectoral Importance

- In China, the manufacturing sector dominates GDP forecasting (driven by its major share in the A-share market)

Sector	Original (US)					Sector	Replication (China)				
	$h = 1$	$h = 2$	$h = 4$	$h = 8$	Average, %		$h = 1$	$h = 2$	$h = 4$	$h = 8$	Average, %
Manufacturing	19.21	24.70	35.43	35.63	28.74	Manufacturing	60.90	61.91	65.58	60.65	62.26
Transportation	52.43	35.12	12.15	7.84	26.89	Information technology	7.09	5.99	5.43	4.46	5.74
Finance	3.78	9.10	23.68	27.30	15.97	Wholesale and retail	6.40	4.20	3.98	6.98	5.39
Services	7.14	7.38	15.03	15.75	11.32	Real estate	3.72	4.85	5.28	2.87	4.18
Retail	5.17	6.09	2.53	4.25	4.51	Transportation & warehousing	2.88	3.41	3.23	6.73	4.06
AR term	7.52	7.36	0.07	0.23	3.80	Electricity, heat, gas & water	2.14	3.69	3.61	2.00	2.86
Wholesale	3.01	5.93	3.34	2.05	3.58	Leasing & business services	1.43	2.88	2.73	1.92	2.24
Mining	1.09	1.73	4.43	4.78	3.01	Mining	1.93	1.79	1.58	3.60	2.22
Construction	0.14	0.29	1.56	1.19	0.79	AR term	4.83	2.02	0.26	0.86	1.99
Agriculture	0.36	2.06	0.38	0.12	0.73	Construction	1.79	1.84	1.83	1.68	1.79
Public administration	0.15	0.23	1.41	0.86	0.66	Culture, sports & entertainment	2.00	1.87	1.01	2.17	1.76

Notes. This table reports the detailed results of sectoral variable importance analysis based on RF

Firm-Level Variable Importance and Firm Characteristics

- In China's bank-dominated financial system, highly leveraged firms are more sensitive to credit cycle fluctuations

Variable	Original (US)				Replication (China)			
	(1) All h	(2) $h = 1$	(3) $h = 2, 4$	(4) $h = 8$	(1) All h	(2) $h = 1$	(3) $h = 2, 4$	(4) $h = 8$
Age	0.228*** (0.0149)	0.386*** (0.0334)	0.160*** (0.0187)	0.102*** (0.0311)	-0.0393 (0.0435)	-0.131 (0.0825)	0.0611 (0.0549)	-0.123 (0.108)
Age squared	-0.00146*** (0.000113)	-0.00226*** (0.000255)	-0.00109*** (0.000141)	-0.000739*** (0.000240)	0.000264 (0.00081)	0.000504 (0.00159)	-0.00059 (0.00102)	0.00147 (0.00197)
Assets (log)	0.0651* (0.0337)	0.425*** (0.0794)	0.0397 (0.0425)	-0.248*** (0.0645)	0.388*** (0.0619)	-0.462*** (0.11)	0.151* (0.0797)	1.73*** (0.156)
Sales growth	-0.338 (0.272)	-0.114 (0.651)	-0.185 (0.339)	-0.928* (0.519)	0.0265 (0.149)	-0.145 (0.27)	0.117 (0.192)	-0.00888 (0.365)
Investment rate	2.66*** (0.601)	1.19 (1.51)	2.04*** (0.749)	4.72*** (1.09)	0.189 (0.165)	1.64*** (0.3)	-0.297 (0.21)	-0.35 (0.414)
BM ratio	-0.363*** (0.0707)	-0.501*** (0.166)	-0.159* (0.0894)	-0.552*** (0.134)	-3.12*** (0.464)	-2.12*** (0.825)	-3.38*** (0.599)	-3.89*** (1.16)
Tobin's q	0.220*** (0.0338)	0.478*** (0.0857)	0.161*** (0.0422)	-0.0378 (0.0609)	-0.195** (0.0798)	0.285** (0.137)	-0.587*** (0.105)	0.0313 (0.199)
Debt/assets	-1.96*** (0.265)	-5.71*** (0.654)	-1.22*** (0.333)	0.578 (0.486)	0.778* (0.466)	2.87*** (0.839)	2.06*** (0.599)	-3.97*** (1.16)
Dummy: paying dividend	2.74*** (0.165)	4.06*** (0.382)	3.09*** (0.212)	-0.0466 (0.311)	0.00538 (0.256)	0.41 (0.461)	-0.581* (0.328)	0.607 (0.646)
Tangible/assets	2.69*** (0.419)	7.24*** (0.972)	1.03* (0.532)	0.275 (0.797)	1.01** (0.478)	-2.71*** (0.859)	0.357 (0.613)	5.7*** (1.2)
Observations	291,458	78,524	151,990	60,944	182,160	46,083	91,809	44,268
R^2	0.021	0.055	0.020	0.004	0.003	0.011	0.005	0.008

Notes. Regressing firm-level relative variable importance on firm characteristics

Extension

- Construct a GDP forecasting model grounded in economic priors
 1. Train an ML model to approximate DSGE-implied GDP forecasts
 2. Fine-tune ML algorithms on empirical data
 3. May outperform both purely theoretical and purely data-driven approaches